A1.1.2 Describe the role of a GPU.

Content for this learning statement

A1.1.2 Describe the role of a GPU.

• The architecture that allows graphics processing units (GPUs) to handle specific tasks and makes them

suitable for complex computations

• Real-world scenarios may include video games, artificial intelligence (AI), large simulations and other

applications that require graphics rendering and machine learning.

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Big Idea: What is a GPU and Why Is It Important?

A **GPU (Graphics Processing Unit)** is a specialized processor designed to perform **parallel processing** of large sets of data, particularly for tasks involving graphics rendering and complex mathematical computations. Originally designed for handling image output to a display, GPUs have become essential components in modern computing systems due to their **highly parallel architecture**.

It would be very smart of you to read and understand how computers process an image. The manner in which images are represented **and then manipulated** make GPU's especially well-suited to working with images.

GPU Architecture and Its Role

At the architectural level, a GPU consists of **hundreds or thousands of smaller cores** that can perform simple calculations **simultaneously**. This makes it ideal for **SIMD (Single Instruction, Multiple Data)** operations — executing the same instruction on many pieces of data at once.

In contrast, a **CPU** (Central Processing Unit) has fewer cores optimized for **sequential** processing and general-purpose tasks. The GPU's architecture allows it to handle **complex visual processing** and **mathematical models** much faster than a CPU could on its own.

- 🔷 **Key Feature:** High-throughput parallelism
- 🔷 **Core Design:** Many smaller, simpler cores
- 🔷 **Best At:** Repetitive tasks across large datasets

Real-World Applications

1. **Video Games**
   GPUs handle 3D rendering, shading, lighting, and motion blur to create realistic graphics in real-time. They process millions of polygons per frame to maintain smooth gameplay.
2. **Artificial Intelligence and Machine Learning**
   Training deep learning models involves **matrix operations** on vast datasets. GPUs accelerate these by performing many operations in parallel, drastically reducing training time.
3. **Large-Scale Simulations**
   Simulations of physical systems (like weather, fluid dynamics, or astrophysics) require massive computation. GPU-based computing clusters are used to model these in a feasible timeframe.
4. **Medical Imaging and Scientific Research**
   GPUs speed up rendering of complex MRI data, protein folding simulations, and other high-performance scientific calculations.

Summary

A GPU is designed to **process large amounts of data in parallel**, making it ideal for tasks like graphics rendering and machine learning. Its architecture — with many simple cores — enables it to outperform CPUs for specific types of computation, especially those involving repeated operations on large data arrays.

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Theme A: Concepts of computer science Computer Fundamentals A1.1 Computer hardware and operation

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